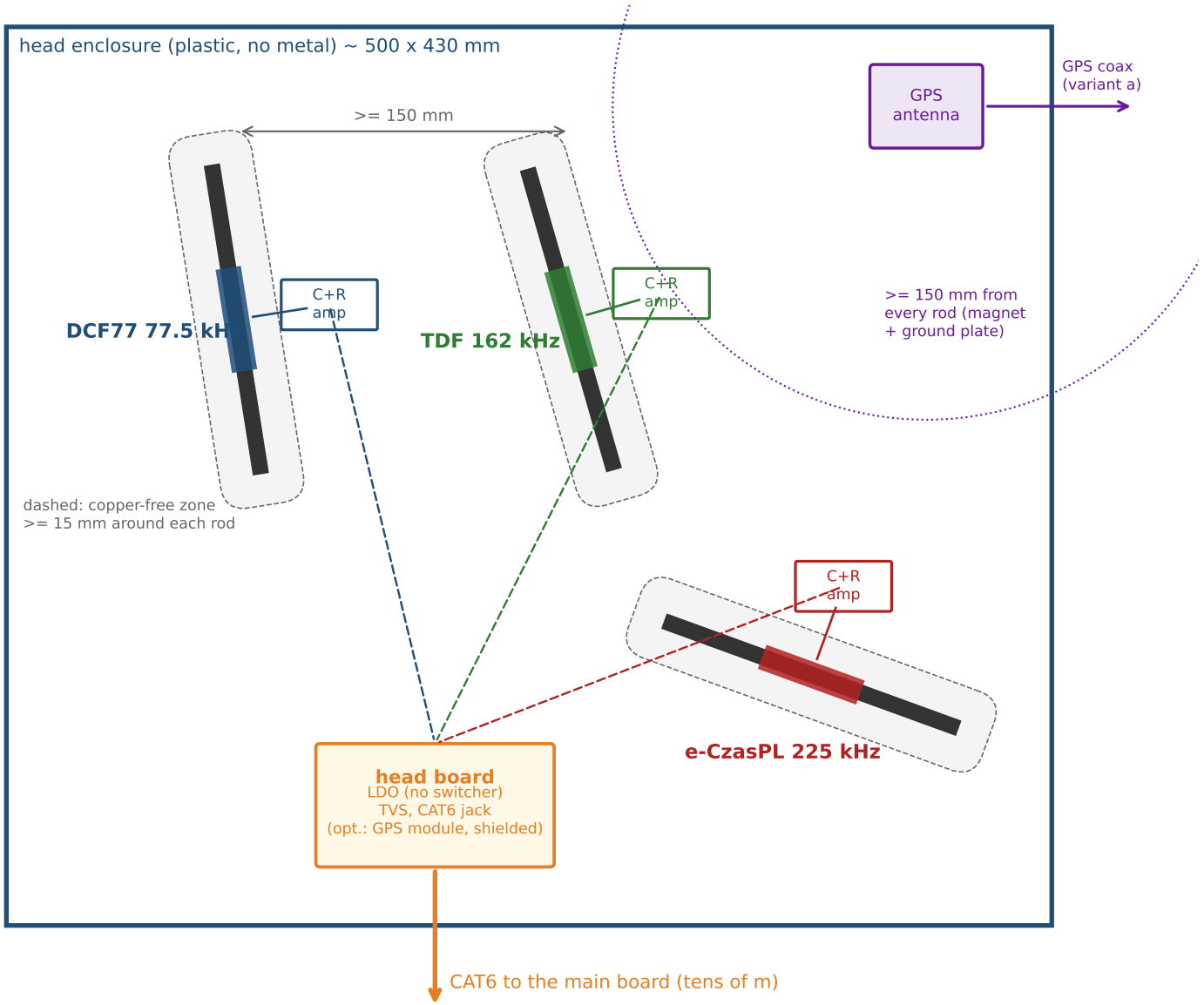
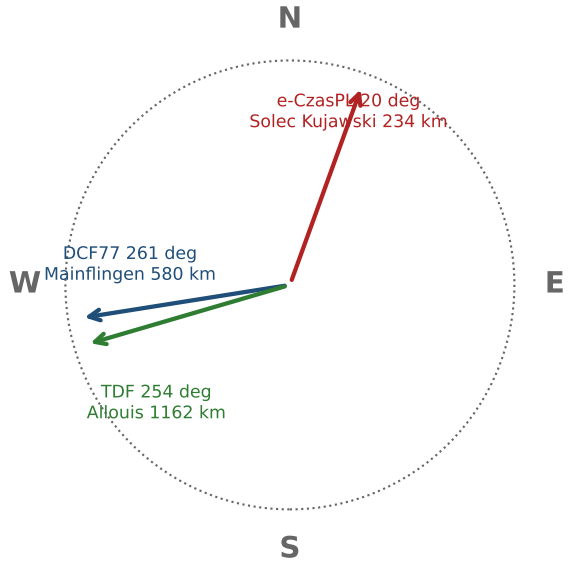
Antenna head - top view to scale (mm), north up

Plastic box with sky view (attic / roof / window). Rods 10 x 150 mm, L1 500 uH + 7-turn L2 (rod-antenna-spec.md); one amplifier module per rod; CAT6 to the main board.



station bearings from Wroclaw



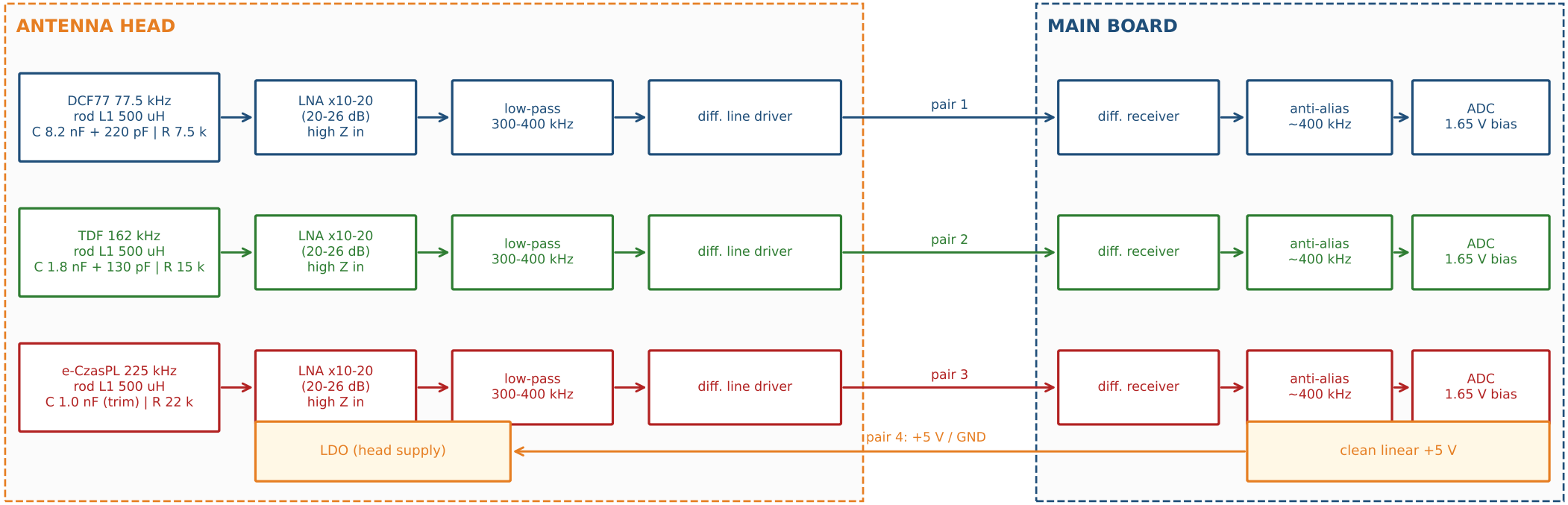
Each rod lies broadside to its station (axis perpendicular to the bearing): DCF77 axis ~171 deg, TDF ~164 deg (nearly parallel, side by side); e-CzasPL ~110 deg.

The head board sits in line with the DCF77 / TDF rod axes: a rod is deaf along its own axis (the null of its figure-8).

Tank (L1 + C + R, Q ~ 30) at each rod; amplifier (20-26 dB) + line driver in the module next to it; tune with the module connected (SPECTRUM hump on the station's frequency).

Install: turn the head so 'north' points north.

Antenna head - signal chain per station, cable and main board



CAT5e / CAT6, differential twisted pairs

pair	variant A (three lines)	variant B (one line + GPS in the head)
1	DCF77	three stations summed
2	TDF	GPS PPS (RS-422)
3	e-CzasPL	GPS UART TX (RS-422)
4	+5 V / GND	+5 V / GND

Values: C0G capacitors at the rod (C = 1/((2 pi f)^2 L), recompute from each rod's measured L), R across L1 for Q ~ 30.

Why the amplifier: the rod's antenna noise must stay well above the ADC noise floor (measured ~15 dB margin at the tank; the 7-turn link alone would cost ~18 dB).

LNA: <= 10 nV/sqrt(Hz), GBW >= 50 MHz for x20 at 225 kHz (or 2 stages).

Variant B: one ADC input for all three stations (STM32H730), separated by the firmware's digital mixers.

ESD/TVS at both cable ends; common-mode rejection by the pairs.